

What is Possible in BRIDGES?

Kalpathi Subramanian¹, Erik Saule¹, Jamie Payton²
krs@charlotte.edu, esaule@charlotte.edu, payton@temple.edu

¹The University of North Carolina at Charlotte

²Temple University

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bridgesuncc.github.io/tutorials/Data_IGN_Games.html

IGN Games Dataset

This dataset contains a collection of game reviews at IGN and is adapted from Reddit User: CDanger (Twitter: @coreyaustir who scraped the data from IGN's website).

Data end point: [IGN Game Link](#)

The program snippet below illustrates how to access the IGN data.

Accessing IGN Games Data: An Example BRIDGES program

Java

C++

Python

```
import java.util.List;
import java.util.Random;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Game;

// this program fragment illustrates how to access the IGN Games data
public class game_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        /* Get a List of Game objects from Bridges */
        DataSource ds = bridges.getDataSource();
        List<Game> mylist = ds.getGameData();

        // Inspect a random Game object
        Game game1 = mylist.get((new Random()).nextInt(mylist.size()));
        System.out.println(game1.getTitle());
        System.out.println(game1.getPlatformType());
        System.out.println(game1.getRating());
        System.out.println(game1.getGenre());
    }
}
```

- Copy/paste the code into your environment and compile.
- Assuming all your code is correct and it compiles correctly, the code will print a record of the data to the console.

bridgesuncc.github.io/doc/java-api/current/ht...

bridges.data_src_dependent.Game Class Reference

Public Member Functions | Package Attributes | List of all members

Detailed Description

A `Game` object, used along with the Games data source.

This is a convenience class provided for users who wish to use this data source as part of their application. It provides an API that makes it easy to access the attributes of this data set.

One would not normally create an object of this type, but rather obtain them through a call to `bridges::connect::DataSource::getGameData()`.

Each game has a title, platform on which it can be played, rating and a list of genres

Refer to tutorial examples on how to use this dataset: https://bridgesuncc.github.io/tutorials/Data_IGN_Games.html

Author

Kalpathi Subramanian

Date

2/1/17, 12/26/20

Public Member Functions

Game ()

Game (String title, String platform, double rating, Vector< String > genre)

String getTitle ()

void setTitle (String title)

String getPlatformType ()

void setPlatformType (String platform)

double getRating ()

void setRating (double rating)

Vector< String > getGenre ()

void setGenre (Vector< String > genre)

Package Attributes

String platform

Vector< String > genre

Constructor & Destructor Documentation

USGS Earthquake Data

This data set is extracted as Tweets from the [US Geological Survey \(USGS\)](#). The BRIDGES API extracts the quake data record periodically and retains the most recent 5000 quakes in a database. These are accessed by the BRIDGES client, reformatted a list of objects for use in assignments.

The program snippet below illustrates how to access the earthquake data.

Accessing Earthquake Data: An Example BRIDGES program

Java

C++

Python

```
import java.util.List;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.EarthquakeUSGS;

// This program fragment illustrates how to access the USGS earthquake data
public class eq_snippet {
    public static void main(String[] args) throws Exception {

        // create Bridges object
        Bridges bridges (YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");
        // read the earth quake data and build the BST
        bridges.setTitle("Accessing USGS Earthquake Data (USGS Data)");

        DataSource ds = bridges.getDataSource();

        // get 100 most recent earthquake records
        List<EarthquakeUSGS> eq_list = ds.getEarthquakeUSGSData(100);

        // print the first record
        System.out.println("Earthquake @:");
        System.out.println("\tMagnitude:" + eq_list.get(0).getMagnitude() + "\n"
            + "\tDate:" + eq_list.get(0).getTime() + "\n"
            + "\tLocation:" + eq_list.get(0).getLocation() + "\n"
            + "\tLat/Long:" + eq_list.get(0).getLatit() + " " +
            eq_list.get(0).getLongit());
    }
}
```

Detailed Description

This class stores the information of an earthquake, retrieved from US Geological Survey Data tweet source.

One would normally not create an object of that type but rather obtain one by calling `bridges::connect::DataSource::getEarthquakeUSGSData()`

Refer to the tutorial on how to use this dataset: https://bridgesuncc.github.io/tutorials/Data_EQ_USGS.html

Author

Mihai Mehedint, Kalpathi Subramanian

Date

12/26/20

Modifications: removed dependencies on Tweet and its ancestors

Public Member Functions

`EarthquakeUSGS ()`

`EarthquakeUSGS (String content, Date date2, double magnitude, double longit, double latit, String location, String title, String url, String properties)`

String `getProperties ()`

void `setProperties (String properties)`

double `getLatit ()`

void `setLatit (double latit)`

double `getLongit ()`

void `setLongit (double longit)`

String `getLocation ()`

void `setLocation (String location)`

String `getTitle ()`

void `setTitle (String title)`

String `getUri ()`

void `setUri (String url)`

void `setMagnitude (double magnitude)`

void `setTime (String t)`

String `getTime ()`

`EarthquakeUSGS (EarthquakeUSGS eq)`

void `eqProperties (String prop)`

void `setMagnitude (Double mag)`

String `enterCarriageReturn (String str)`

double `getMagnitude ()`

Bridges - Elevation Tutorial - Mozilla Firefox

Bridges - Elevation Tutorial x +

bridgesuncc.github.io/tutorials/Data_Elevation.ht

Step 1: Suggested Examples:

data_source.get_elevation_data([36.8241, -116.8369, 35.7797, -117.4074])
data_source.get_elevation_data([49.0872, -113.6184, 47.3746, -114.8143])
data_source.get_elevation_data([40.1924, -79.3980, 37.5021, -81.8810])

Accessing Elevation Data: An Example BRIDGES program

Java C++ Python

import java.util.List;
import java.util.Random;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.ElevationData;

// a program fragment to access the Elevation data
public class elevation_snippet {
 public static void main(String[] args) throws Exception {

 //create the Bridges object
 Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

 // Get a List of Elevation data records from Bridges
 DataSource ds = bridges.getDataSource();
 ElevationData elev_data = ds.getElevationData(41.33133177632377,
 -98.02593749997456, 42.508577297430456, -96.94531249997696, 0.02);

 System.out.println("Width: " + elev_data.getCols());
 System.out.println("Height: " + elev_data.getRows());
 System.out.println("Resolution: " + elev_data.getCellSize());
 System.out.println("Lower Left Corner: " + elev_data.getxll() + ", " + elev_data.getyll());

 System.out.println("first 10 elevations:");
 for (int i = 0; i < 10; ++i) {
 System.out.println(elev_data.getData()[i][1]);
 }
 }
}

Bridges-Java: bridges.data_src_dependent.ElevationData Class Reference - Mozilla Firefox

Bridges-Java: bridges.data_src_dependent.ElevationData x +

https://bridgesuncc.github.io/doc/java-api/current/ht

Public Member Functions | List of all members

bridges.data_src_dependent.ElevationData

Class Reference

Detailed Description

Object that holds elevation data retrieved from NOAA repository.

A user would not normally create an `ElevationData` object but rather obtain one from calling `bridges::connect::DataSource::getElevationData()`

A tutorial on how to use the Elevation dataset is available at: https://bridgesuncc.github.io/tutorials/Data_Elevation.html

Author

Jay Strahler

Date

12/26/20

Public Member Functions

ElevationData ()
ElevationData (int[][] data, int cols, int rows, double xll, double yll, double cellsize, int maxVal, int minVal)
int[][] getData ()
int getVal (int r, int c)
void setData (int[][] data)
int getCols ()
void setCols (int cols)
int getRows ()
void setRows (int rows)
double getxll ()
void setxll (double xll)
double getyll ()
void setyll (double yll)
double getCellSize ()
void setCellSize (double cellsize)
int getMaxVal ()
void setMaxVal (int maxVal)
int getMinVal ()
void setMinVal (int minVal)

Bridges - Lyrics Tutorial — Mozilla Firefox

bridgesuncc.github.io/tutorials/Data_Song_Lyrics.html

```
ds.get_song("Enter Sandman", "Metallica")
ds.get_song("Around the World", "Daft Punk")
ds.get_song("Never Gonna Give You Up", "Rick Astley")
```

Accessing Song Lyrics Data: An Example BRIDGES program

Java C++ Python

```
import java.util.List;
import java.util.Random;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Song;
import java.util.ArrayList;

// a program fragment illustrating how to access the song lyrics data set

public class song_lyrics_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        // set title
        bridges.setTitle("Accessing Song Data");

        // create data source object
        DataSource ds = bridges.getDataSource();

        // Get Song data
        Song s = ds.getSong("Harder Faster Better Stronger", "Daft Punk");

        // print song attributes
        System.out.println(s.getSongTitle());
        System.out.println(s.getArtist());
        System.out.println(s.getAlbumTitle());
        System.out.println(s.getReleaseDate());
        System.out.println(s.getLyrics());
    }
}
```

Bridges-Java: bridges.data_src_dependent.Song Class Reference — Mozilla Firefox

bridgesuncc.github.io/doc/java-api/current/html

Main Page Packages Classes Files

bridges data_src_dependent Song

Public Member Functions | Package Attributes | List of all members

bridges.data_src_dependent.Song Class Reference

Detailed Description

A **Song** object, used along with the Songs data source (using the Genius API).

This is a convenience class provided for users who wish to use this data source as part of their application. It provides an API that makes it easy to access the attributes of this data set. The **Song** object is typically obtained from calling `bridges::connect::DataSource::getSongData()` or `bridges::connect::DataSource::getSong()`.

Refer to tutorial for example of using this feature: https://bridgesuncc.github.io/tutorials/Data_Song_Lyrics.html

Author
David Burlinson

Date
5/21/18

Public Member Functions

Song ()
Song (String artist, String song, String album, String lyrics, String release_date)
String getArtist ()
void setArtist (String artist)
String getSongTitle ()
void setSongTitle (String song)
String getAlbumTitle ()
void setAlbumTitle (String album)
String getLyrics ()
void setLyrics (String lyrics)
String getReleaseDate ()
void setReleaseDate (String release_date)

Package Attributes

String song

Bridges - Gutenberg Tutorial — Mozilla Firefox

Bridges - Gutenberg Tutorial

Step 1: Suggested Examples:

```
data_source.get_gutenberg_book_metadata("Shakespeare, William", "author")
data_source.gutenberg_book_text(2701)
```

The program snippet below illustrates how to access this dataset.

Accessing Gutenberg Data: An Example BRIDGES program

Java C++ Python

```
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Amenities;
import bridges.data_src_dependent.*;
import java.util.*;

public class gutenberg_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(0, "YOUR_USER_ID", "YOUR_API_KEY");

        //create a bridges datasource object to use and
        //retrieve the gutenberg books
        DataSource ds = bridges.getDataSource();

        //Get Meta data for a single book

        //Get Meta data for a book

        System.out.println("Meta data for 1 book: Moby Dick by Id");
        GutenbergMeta b = ds.getAGutenbergBookMetaData(2701);
        System.out.println("\tId: " + b.getId());
        System.out.println("\tTitle: " + b.getTitle());
        System.out.println("\tLanguage: " + b.getLanguage());
        System.out.println("\tDate Added: " + b.getDate());

        System.out.println("Retrieving books by title: Pride and Prejudice");
        List<GutenbergMeta> blist = ds.getGutenbergBookMetaData("Pride and Prejudice", "title");
```

Bridges-Java: bridges.data_src_dependent.GutenbergMeta Class Reference — Mozilla Firefox

Bridges-Java: bridges.data_src_dependent.GutenbergMeta Class Reference

Detailed Description

Stores the meta data of a book as stored by project Gutenberg.

This object stores the meta data of a book such as its title, author and genre. The meta data come from <https://www.gutenberg.org/>.

Objects of this type are typically not constructed by the user but returned by a call to our Gutenberg API such as `bridges::connect::DataSource::getGutenbergBookMetaData()` or `bridges::connect::DataSource::getAGutenbergBookMetaData(int)`.

The object does not contain the text of the book itself. Though it can be obtained using `bridges::connect::DataSource::getGutenbergBookText()`

A tutorial of how to use the Gutenberg data in BRIDGES is available: https://bridgesuncc.github.io/tutorials/Data_Gutenberg.html

Public Member Functions

<code>GutenbergMeta ()</code>	
<code>GutenbergMeta (int id, String title, String lang, String date, String[] authors, String[] genres, String[] loc)</code>	
<code>int getId ()</code>	GutenbergID of the book. More...
<code>void setId (int id)</code>	
<code>String getTitle ()</code>	Title of the book. More...
<code>void setTitle (String title)</code>	
<code>String getLanguage ()</code>	language of the book. More...
<code>void setLanguage (String lang)</code>	
<code>String getDate ()</code>	date at which the book was added to project Gutenberg. More...
<code>void setDate (String date)</code>	
<code>String[] getAuthors ()</code>	List of authors. More...
<code>void setAuthors (String[] authors)</code>	
<code>String[] getGenres ()</code>	List of genres. More...
<code>void setGenres (String[] genres)</code>	
<code>String[] getLoc ()</code>	List of library of congress classifications. More...

- **BRIDGES Initialization, Credentials:** This is typically done **once** at the beginning of the application. Credentials include your user id and application authentication id (available via your BRIDGES login). Here this is required to access the DataSource object which controls all calls to external datasets.
- **Retrieving the data** This is a single call to a method in DataSource to return object(s) of that data type, and may include parameters.
- **Printing an example record** The program will print an example record of this dataset, to illustrate its attributes.

java

C++

Python

```
import java.util.List;
import java.util.Random;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Shakespeare;

// a program fragment to access the Shakespeare data (plays, poems)

public class shakespeare_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        // Get a List of Shakespeare objects from Bridges
        DataSource ds = bridges.getDataSource();
        List<Shakespeare> mylist = ds.getShakespeareData();

        // Inspect a random Shakespeare object
        Shakespeare work1 = mylist.get((new Random()).nextInt(mylist.size()));
        System.out.println(work1.getTitle());
        System.out.println(work1.getType());
        System.out.println(work1.getText().substring(0, Math.min(100, work1.getText().length())) + "...");
    }
}
```

- Copy/paste the code into your environment and compile.
- Assuming all your code is correct and it compiles correctly, the code will print a record of the data to the console.

bridges.data_src_dependent.Shakespeare Class Reference

[Public Member Functions](#) | [Package Attributes](#) | [List of all members](#)

Detailed Description

A [Shakespeare](#) Data source object containing sonnets, poems and plays.

This is a convenience class provided for users who wish to use this data source as part of their application. It provides an API that makes it easy to access the attributes of this data set.

One would not normally create an object of this type but rather get one by calling `bridges::connect::DataSource::getShakespeareData()`

Refer to the tutorial on how to use this dataset: https://bridgesuncc.github.io/tutorials/Data_Shakespeare.html

Author

Kalpathi Subramanian

Date _____

2/1/17, 12/26/20

Public Member Functions

Shakespeare ()

Shakespeare (String title, String type, String text)

```
String getTitle ()
```

```
void setTitle (String title)
```

String getType()

```
void setType (String type)
```

String getText()

```
void setText (String text)
```

Package Attributes

String type

String text

Constructor & Destructor Documentation

- ◆ Shakespeare() [1/2]

Bridges - IMDb Tutorial — Mozilla Firefox

Bridges - IMDb Tutorial × +

← → ↺ 🔒 bridgesuncc.github.io/tutorials/Data_IMDB.html ☆ 📧 ⚙️ 🔍 🔄 🌐 📄 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📔 📕 📖 📗 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿

IMDB Actor/Movie Data

This data set is a curated subset of IMDB actor/movie dataset, with each record representing an actor name and a movie in which the actor appeared. There are 1814 actor/movie pairs in this dataset; this dataset is particularly suite to solving the number problem, using graphs and Breadth First Search (BFS).

The program snippet below illustrates how to access this dataset.

Accessing IMDB Data: An Example BRIDGES program

Java C++ Python

```
import java.util.List;
import java.util.Random;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.ActorMovieIMDB;

// A test class that fetches a random movie from IMDB then sends the
// actors name and movie title + release date to the console as output.
// There is no visual output to the users BRIDGES gallery after running this code.

public class imdb_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        DataSource ds = bridges.getDataSource();

        // Get a List of ActorMovieIMDB objects from Bridges
        List<ActorMovieIMDB> mylist = ds.getActorMovieIMDBData(1813);

        // Inspect a random ActorMovieIMDB object
        ActorMovieIMDB pair1 = mylist.get(new Random().nextInt(mylist.size()));
        System.out.println(pair1.getActor());
        System.out.println(pair1.getMovie());
    }
}
```

- Copy/past the code into your environment and compile.

Bridges-Java: bridges.data_src_dependent.ActorMovieIMDB Class Reference — Mozilla Firefox

Bridges-Java: bridges.data_src_dependent.ActorMovieIMDB Class Reference × +

← → ↺ 🔒 https://bridgesuncc.github.io/doc/java-api/current/html/ ☆ 📧 ⚙️ 🔍 🔄 🌐 📄 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📔 📕 📖 📗 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿

bridges.data_src_dependent.ActorMovieIMDB

Class Reference

Detailed Description

This class represents an actor-movie pair object, to be used with the IMDB actor-movie dataset.

A user typically does not create an `ActorMovieIMDB` object. They are usually returned by `bridges::connect::DataSource::getActorMovieIMDBData()` and `bridges::connect::DataSource::getActorMovieIMDBData2()`

Check out the tutorial on the IMDB dataset at https://bridgesuncc.github.io/tutorials/Data_IMDB.html

Author

Kalpathi Subramanian

Public Member Functions

	ActorMovieIMDB ()
String	getActor () get the actor from this object More...
void	setActor (String a) set the actor from this object More...
String	getMovie () get the movie from this object More...
void	setMovie (String m) set the movie from this object More...
double	getMovieRating () get the movie rating from this object More...
void	setMovieRating (double r) set the movie rating for this object More...
Vector<String>	getGenres () get the genres of this movie More...
void	setGenres (Vector<String> g) set the genres for this movie More...

Package Attributes

String movie

Bridges - Wiki Tutorial — Mozilla Firefox

Bridges - Wiki Tutorial × +

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bridgesuncc.github.io/tutorials/Data_WikiDataActor.html

The program snippet below illustrates how to access the Wikidata actor/movie data using the Wikidata source.

Java C++ Python

```
/**
 * Created by Lucas Estrella on 1/31/2017.
 * lestrell@uncc.edu
 */
import java.util.ArrayList;

import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.ActorMovieWikidata;

// This program fragment illustrates how to access and read the Wikidata actor/movie data

public class wikidata_actor_snippet {

    public static void main(String[] args) throws Exception {

        // note: you must fill in with your Bridges credentials
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID",
            "YOUR_API_KEY");
        // set title
        bridges.setTitle("Accessing Wikidata Movie/Actor Data");

        // create data source object
        DataSource ds = bridges.getDataSource();

        // Get a List of Wikidata actor/movie records from Bridges
        // get the actor movie Wikidata data through the BRIDGES API for 1955.
        // data are available from the early 20th century to now.
        ArrayList<ActorMovieWikidata> list = ds.getWikidataActorMovie(1955, 1955);

        System.out.println("Data Records in 1955: " + list.size() + "\n");

        // print out the first 3 records of the dataset
        for (int k = 0; k < 3; k++)
            System.out.println("Actor-Movie Data:\n" +
                "\tMovie: " + list.get(k).getMovieURI() + " \t" + list.get(k).getMovieName() + "\n" +
                "\tActor: " + list.get(k).getActorURI() + " \t" + list.get(k).getActorName() + "\n");
    }
}
```

Bridges-Java: bridges.data_src_dependent.ActorMovieWikidata Class Reference — Mozilla Firefox

Bridges-Java: bridges.data_src_dependent.ActorMovieWikidata Class Reference × +

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https://bridgesuncc.github.io/doc/java-api/current/html

bridges.data_src_dependent.ActorMovieWikidata Class Reference

Detailed Description

Represent Actor Movie relations extracted from Wikidata.

Note that the end user will not create an object of that type in regular circumstances. But rather, **ActorMovieWikidata** objects are returned by `bridges::connect::DataSource::getWikidataActorMovie()`

See also

Example of how to access that type of data is provided at: https://bridgesuncc.github.io/tutorials/Data_WikiDataActor.html

Author

Erik Saule

Public Member Functions

ActorMovieWikidata (String movieURI, String actorURI, String movieName, String actorName)
String getMovieURI () get movie URI More...
void setMovieURI (String movieURI) set movie URI More...
String getActorURI () get actor URI More...
void setActorURI (String actorURI) set actor URI More...
String getMovieName () get movie name More...
void setMovieName (String movieName) set movie name More...
String getActorName () get actor name More...
void setActorName (String actorName) set actor name More...

Package Attributes

String actorURI
String movieName
String actorName

Bridges - OSM Tutorial

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← → ↺ 🔒 bridgesuncc.github.io/tutorials/Data_OSM.html ☆ 📄 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📔 📕 📖 📗 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📿

Accessing Open Street Map Data: An Example BRIDGES program

Java

C++

Python

```
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.OsmData;
import bridges.data_src_dependent.OsmVertex;
import bridges.data_src_dependent.OsmEdge;

// An example program fragment to illustrate how to retrieve Open Street map data.
// The program retrieves the map of UNC Charlotte campus, prints the number of vertices
// and edges, and the location of the first vertex

public class osm_snippet {
    public static void main(String[] args) throws Exception {

        //create the Bridges object
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        DataSource ds = bridges.getDataSource();
        // OsmData osm_data = ds.getOsmData("Charlotte, North Carolina", "default");
        OsmData osm_data = ds.getOsmData(41.83133177632377, -98.82593749997456,
            42.008577297430456, -97.94531249997696);

        OsmVertex[] vertices = osm_data.getVertices();
        OsmEdge[] edges = osm_data.getEdges();

        System.out.println("Number of Vertices [Charlotte]:" + vertices.length);
        System.out.println("Number of Edges [Charlotte]:" + edges.length);

        // get cartesian coordinate location of first vertex
        double[] coords = osm_data.getVertices()[0].getCartesian_coord();
        System.out.println ("Location of first vertex [Cartesian Coord]: " + coords[0] + " " +
            + coords[1]);
    }
}
```

Bridges-Java: bridges.data_src_dependent.OsmData Class Reference

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bridges.data_src_dependent.OsmData Class Reference

Detailed Description

Class that hold Open Street Map vertices.

This class holds Open Street Map data, from <https://openstreetmap.org>

Objects from this class are typically not created by the user but retruned by `bridges::connect::DataSource::getOsmData()`

Check out how to use OSM data at: https://bridgesuncc.github.io/tutorials/Data_OSM.html

Author
Kalpathi Subramanian, Erik Saule

Date
2/16/19, 12/26/20

Public Member Functions

	OsmData ()
	OsmData (OsmVertex[] vertices, OsmEdge[] edges, String name)
OsmVertex[]	getVertices ()
void	setVertices (OsmVertex[] vertices)
String	getName ()
void	setName (String name)
OsmEdge[]	getEdges ()
void	setEdges (OsmEdge[] edges)
double[]	getCartesianRangeY ()
double[]	getLatitudeRange ()
double[]	getLongitudeRange ()
double[]	getCartesianRangeX ()
void	getLatLongRange (double[] latr, double[] lonr)
GraphAdjList< Integer, OsmVertex, Double >	getGraph ()

Package Attributes

double[]	longitude_range
double[]	cartesian_range_x
double[]	cartesian_range_y

Bridges - Amenity Tutorial — Mozilla Firefox

Bridges - Amenity Tutorial x +

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Step 1: Suggested Examples:

```
data_source.get_amenity_data("New York, New York", "food")
data_source.get_amenity_data("Los Angeles, California", "school")
data_source.get_amenity_data("Chicago, Illinois", "firestation")
```

The program snippet below illustrates how to obtain map data of the UNC Charlotte.

Accessing Open Street Map Amenity Data: An Example BRIDGES program

Java C++ Python

```
import java.lang.String;
import bridges.base.SElement;
import bridges.base.GraphAdjList;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Amenities;
import bridges.data_src_dependent.AmenityData;

public class osm_amenity_snippet {
    public static void main(String[] args) throws Exception {

        // initialize Bridges
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID",
            "YOUR_API_KEY");

        // title, description
        bridges.setTitle("An Open Street Map Example");

        // get the OSM data
        DataSource ds = bridges.getDataSource();
        AmenityData amenity_data = ds.getAmenityData(38.77657, -77.20918, 39.03198, -76.8999, "food");

        System.out.println("Total Points: " + amenity_data.getCount());
        System.out.println("ID: " + amenity_data.getAmenities().get(0).getId());
        System.out.println("Name: " + amenity_data.getAmenities().get(0).getName());
        System.out.println("Lat: " + amenity_data.getAmenities().get(0).getLat());
        System.out.println("Lon: " + amenity_data.getAmenities().get(0).getLon());

    }
}
```

Bridges-Java: bridges.data_src_dependent.Amenities Class Reference — Mozilla Firefox

Bridges-Java: bridges.data_src_dependent.Amenities Class Reference x +

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Public Member Functions | List of all members

bridges.data_src_dependent.Amenities Class Reference

Detailed Description

Class that hold individual Open Street Map [Amenities](#).

This class holds the individual information for each amenity requested.

Usually this class is not created by the user but part of an [AmenityData](#) object returned by `bridges::connect::DataSource::getAmenityData()`

Check out the tutorial on getting amenity data at https://bridgesuncc.github.io/tutorials/Data_Amenity.html

Author

Jay Strahler

Public Member Functions

Amenities ()

Amenities (double id_val, double lat, double lon, String name, String[] other)

double getId ()

void setId (double id_val)

double getLat ()

void setLat (double lat)

double getLon ()

void setLon (double lon)

String getName ()

void setName (String name)

String[] getOther ()

void setOther (String[] other)

Constructor & Destructor Documentation

◆ Amenities() [1/2]

bridges.data_src_dependent.Amenities.Amenities ()

Default Constructor

Reddit

```
import java.util.ArrayList;
import bridges.connect.Bridges;
import bridges.connect.DataSource;
import bridges.data_src_dependent.Reddit;

// This program fragment illustrates how to access the reddit data
public class reddit_snippet {
    public static void main(String[] args) throws Exception {

        // create Bridges object
        Bridges bridges = new Bridges (YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID", "YOUR_API_KEY");

        DataSource ds = bridges.getDataSource();

        ArrayList<String> subreddits = ds.getAvailableSubreddits();

        // get some data from the "Ask Science" subreddit
        ArrayList<Reddit> reddit_posts = ds.getRedditData("askscience");
        for (Reddit r: reddit_posts) {
            System.out.println("ID: "+r.getID());
            System.out.println("Title: "+r.getTitle());
            System.out.println("Author: "+r.getAuthor());
            System.out.println("Score: "+r.getScore());
            System.out.println("VoteRatio: "+r.getVoteRatio());
            System.out.println("CommentCount: "+r.getCommentCount());
            System.out.println("Subreddit: "+r.getSubreddit());
            System.out.println("PostTime: "+r.getPostTime());
            System.out.println("URL: "+r.getURL());
            System.out.println("Text: "+r.getText());
            System.out.println("");
        }
    }
}
```

Detailed Description

A reddit object representing a reddit post, used along with the reddit data source.

This is a convenience class provided for users who wish to use this data source as part of their application. It provides an API that makes it easy to access the attributes of this data set.

This object is generally not created by the user, to see how its created check out `bridges::data_src_dependent::data_source::reddit_data()`

For an example, check out https://bridgeunoc.github.io/tutorials/Data_Reddit.html

Author

Erik Saule, Jay Strahler

Date

12/12/22

Public Member Functions

```
def __init__(self, id=0, title="", author="", score=0, vote_ratio=0.0, comment_count=0, subreddit="", post_time=0, url="", text="")
def id(self)
    return id of the reddit post More...
def id(self, i)
def title(self)
    return the title of the reddit post More...
def title(self, i)
def author(self)
    return the author of the reddit post More...
def author(self, i)
def score(self)
    return the score (upvotes-downvotes) of the reddit post More...
def score(self, i)
def vote_ratio(self)
    ratio of upvote to downvotes of the reddit post More...
def vote_ratio(self, i)
def comment_count(self)
    number of comments on that reddit post More...
def comment_count(self, i)
def subreddit(self)
    name of the subreddit the post appeared in More...
def subreddit(self, i)
def post_time(self)
    time the post was made (UNIX time) More...
def post_time(self, i)
def url(self)
    URL associated with the post. More...
def url(self, i)
def text(self)
    returns the text of the reddit post. More...
def text(self, i)
```

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Arrays

```
import bridges.base.ArrayID;

public class arrayId {
    public static void main(String[] args) throws Exception {

        //create the Bridges object, set credentials
        Bridges bridges = new Bridges(YOUR_ASSIGNMENT_NUMBER, "YOUR_USER_ID",
            "YOUR_API_KEY");

        // title, description
        bridges.setTitle("One-Dimensional Array Example");
        bridges.setDescription("One dimensional array with visual attributes");

        // set array dimensions, allocate array of elements
        int arraySize = 10;

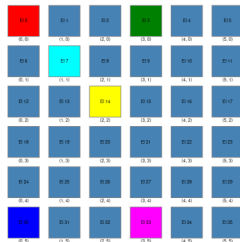
        ArrayID<Integer> arr = new ArrayID<Integer> (arraySize);

        // populate the array, with squares of indices
        // use the values to label the elements
        for (int k = 0; k < arr.getSize(); k++) {
            arr.getElement(k).setValue(k * k);
            arr.getElement(k).setLabel(String.valueOf(arr.getElement(k).getValue()));
        }

        // color the array elements
        arr.getElement(0).setColor("red");
        arr.getElement(1).setColor("green");
        arr.getElement(2).setColor("blue");
        arr.getElement(3).setColor("cyan");
        arr.getElement(4).setColor("magenta");
        arr.getElement(5).setColor("yellow");
        arr.getElement(6).setColor("red");
        arr.getElement(7).setColor("green");
        arr.getElement(8).setColor("blue");
        arr.getElement(9).setColor("black");

        // tell Bridges what datastructure to visualize
        bridges.setDataStructure(arr);

        // visualize the list
        bridges.visualize();
    }
}
```



(Slice: 0)



(Slice: 1)



(Slice: 2)

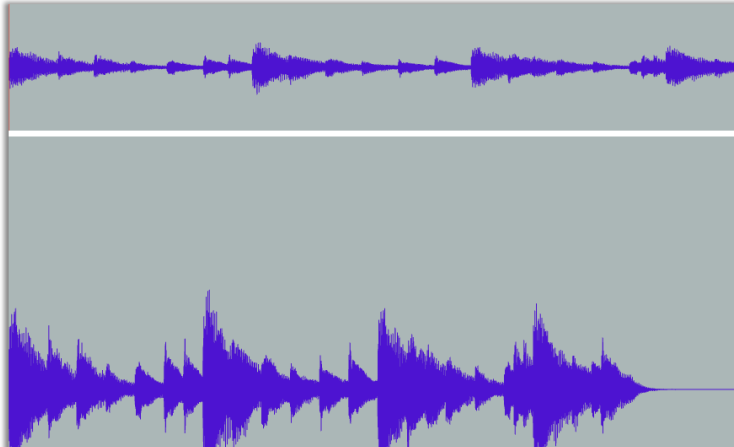


AudioClip

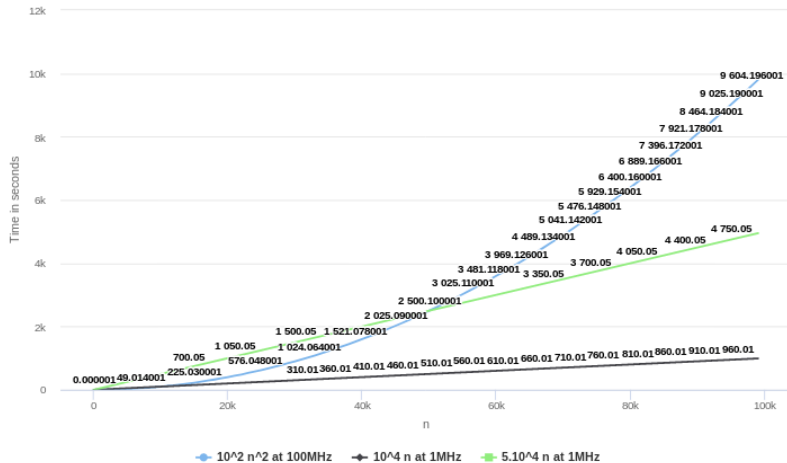
Audio Mixing

Mixing two AudioClip objects together

33.00



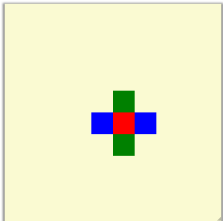
LineChart



ColorGrid

BRIDGES Color Grid Tutorial - Part 1

This is an example of the color grid with height and width of 10 units each.



null

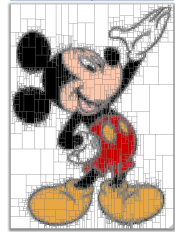
null



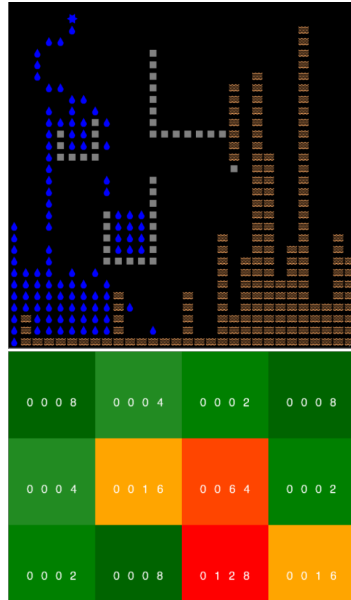
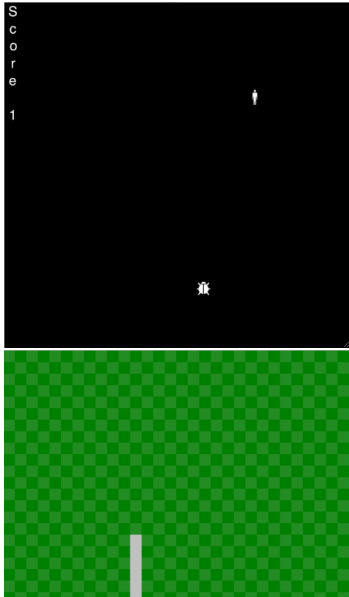
KD Tree Representation of an Image

null

null



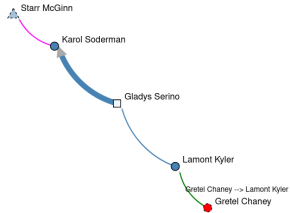
Games



Linked List

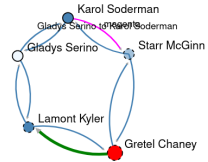
A Singly Linked List Example

Demonstrate styling linked list nodes and links



A Circular Doubly Linked List Example

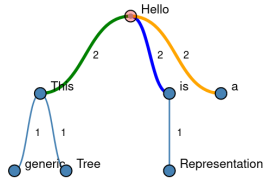
This example shows circular doubly linked example with five nodes with assigned visual attributes



Trees

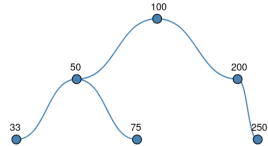
A General Tree Example

A basic tree with seven nodes. Three on one side and three on the other. The root node is set to red with 0.3 opacity. The other six nodes are neutral color.



A Simple Binary Search Tree Example

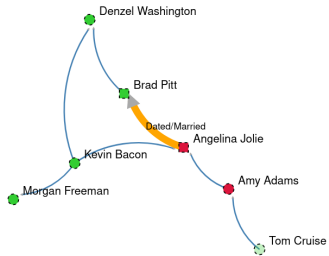
This example illustrates a binary search tree built using BRIDGES



Graphs

A Simple Adjacency list based Graph Example.

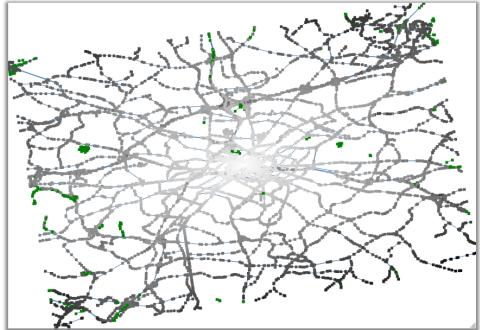
Demonstrate styling graph nodes and links with visual attributes



Graph : OpenStreet Map Example

Charlotte: Color of vertices is based on distance from center of the city

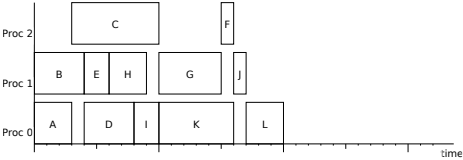
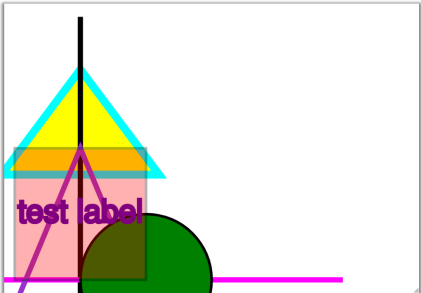
9.00



Shape Collection

Symbol Collection

Red square, green circle, magenta horizontal and vertical lines, and a test label with a purple outline.



PQBook

MinHeap represented as a Binary Tree.

8:00

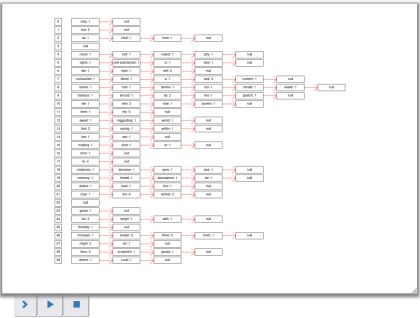


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Assignment Repository

- Structured by course and topics: [\[BRIDGES Assignments\]](#)
- By course/learning outcomes: [\[PDF\]](#)